

Answer key

The coordinate grid. Every point on the grid has two numbers: its x (how far across) and its y (how far up), written (x, y) . The middle, where the lines cross, is the **origin** $(0, 0)$. Right and up are positive; left and down are negative. A *turtle* sits on the grid and follows instructions to move around and draw.

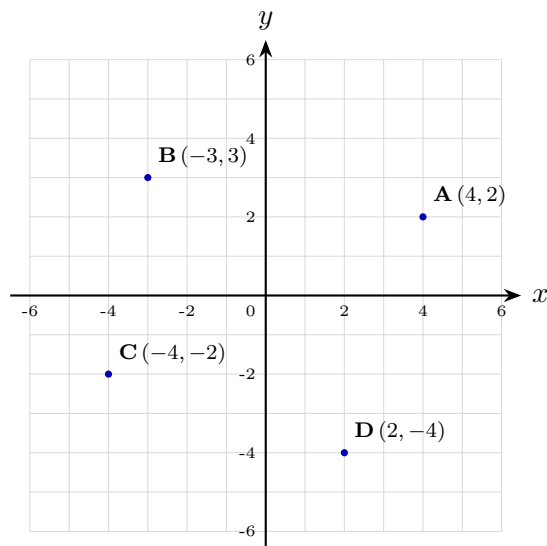
This is exactly how Scratch works. The Scratch stage is one big coordinate grid with $(0, 0)$ in the middle: x runs from -240 on the far left to 240 on the far right, and y from -180 at the bottom to 180 at the top. The blocks go to x : y :, change x by, change y by and move/turn move a sprite around that grid — the very instructions you practise below.

Command cheat sheet. Write every answer using *only* these instructions — they are the real Scratch blocks. Fill the () slots with numbers.

go to x: (x) y: (y)	jump straight to the point (x, y)
change x by (n)	move across: right is +, left is –
change y by (n)	move up or down: up is +, down is –
move (n) steps	go forward in the direction you are facing
turn right (n) degrees	turn clockwise (changes facing, draws nothing)
turn left (n) degrees	turn anticlockwise (changes facing, draws nothing)
pen up	lift the pen: move without drawing
pen down	put the pen down: draw as you move

Problem 1.

Read the grid. Write the coordinates (x, y) of each labelled point.

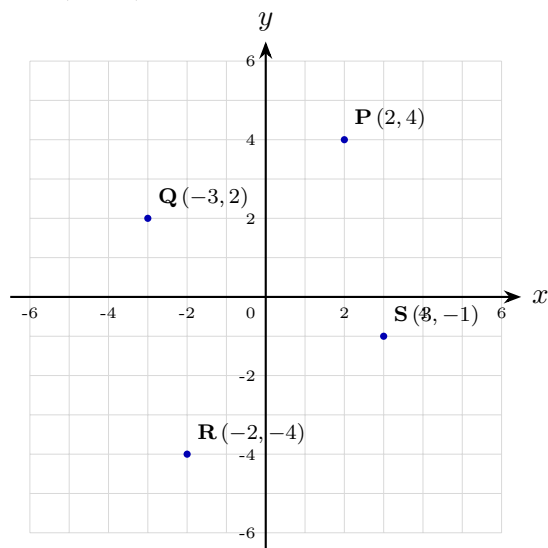


$$A = (4, 2) \quad B = (-3, 3) \quad C = (-4, -2) \quad D = (2, -4)$$

Notice the signs: top-right is $(+, +)$, top-left $(-, +)$, bottom-left $(-, -)$, bottom-right $(+, -)$. Scratch points work the same way.

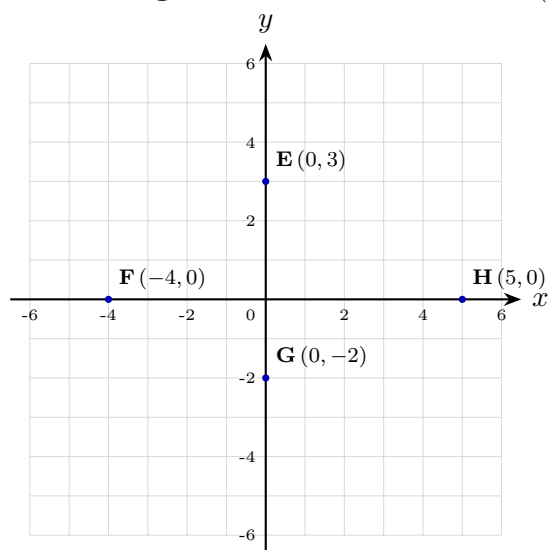
Problem 2.

Plot the points. Mark and label these points on the grid: $\mathbf{P} = (2, 4)$, $\mathbf{Q} = (-3, 2)$, $\mathbf{R} = (-2, -4)$, $\mathbf{S} = (3, -1)$.



Problem 3.

Read the grid. Write the coordinates (x, y) of each labelled point.



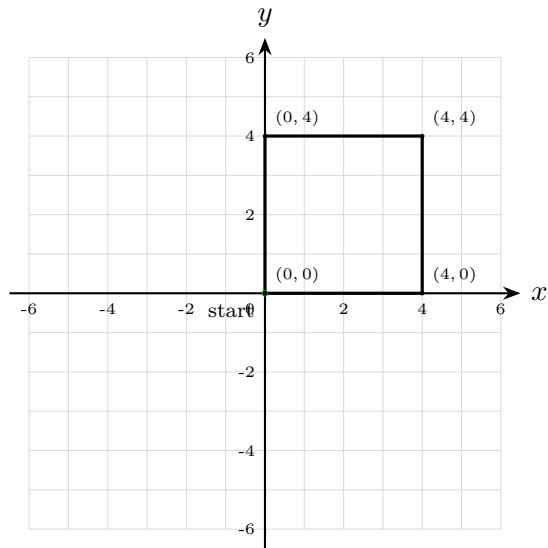
$$\mathbf{E} = (0, 3) \quad \mathbf{F} = (-4, 0) \quad \mathbf{G} = (0, -2) \quad \mathbf{H} = (5, 0)$$

These points sit on a line, not in a corner — so one of their two numbers is 0.

A point on the up-down line has $x = 0$; a point on the across line has $y = 0$. The origin $(0, 0)$ is on both.

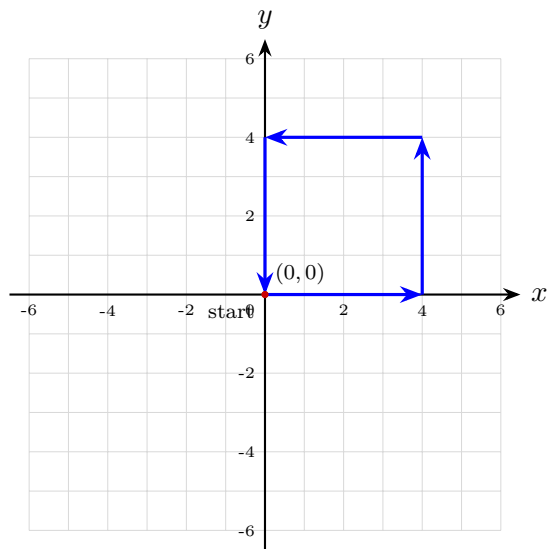
Problem 4.

Instruct the turtle. The turtle starts at the origin with its pen down. Write `go to` instructions to draw this square, visiting each corner in order.



One solution:

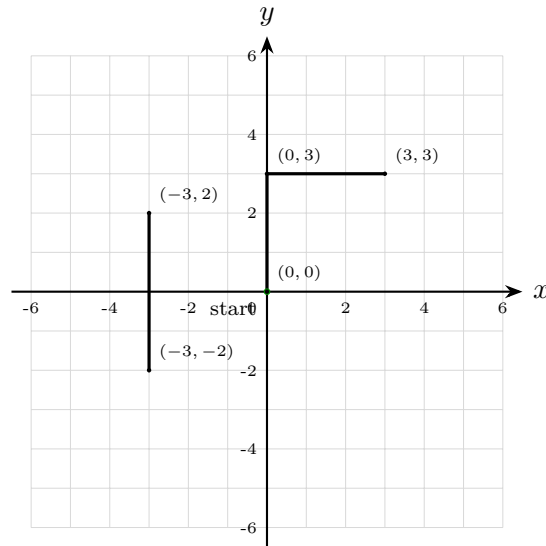
```
go to x: 4 y: 0  
go to x: 4 y: 4  
go to x: 0 y: 4  
go to x: 0 y: 0
```



Problem 5.

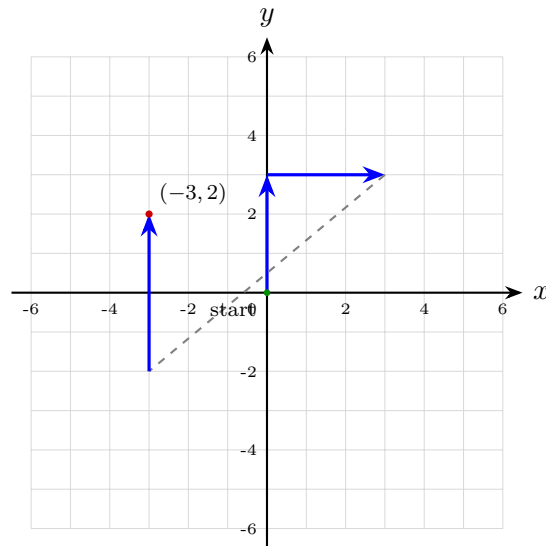
Instruct the turtle. Draw these two separate strokes. The turtle's pen must be lifted while it jumps from the end of the first stroke to the start of the second.

Use *pen up* before the jump and *pen down* after it, or the jump will leave an unwanted line.



One solution:

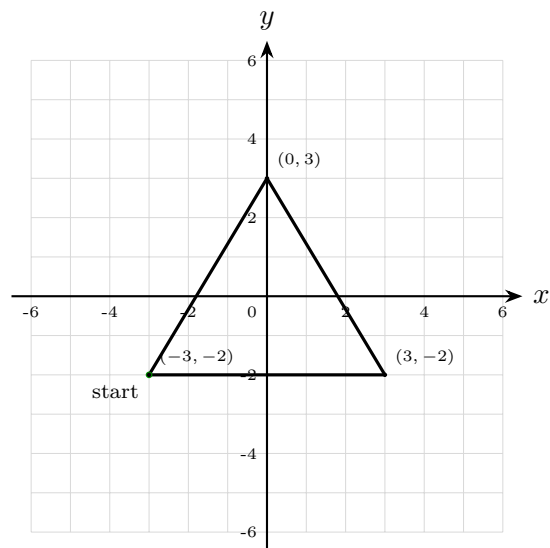
```
go to x: 0 y: 3
go to x: 3 y: 3
pen up
go to x: -3 y: -2
pen down
go to x: -3 y: 2
```



A dashed line in the answer key means the pen was up — the turtle moved but drew nothing.

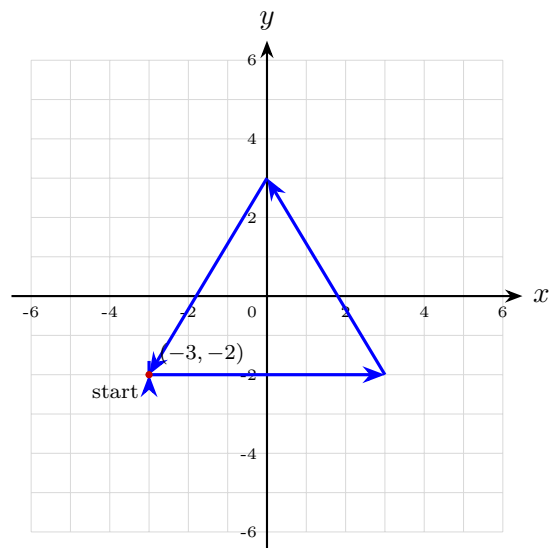
Problem 6.

Instruct the turtle. Starting at the bottom-left spot, write `go to` instructions to visit the three treasure spots and return to the start, drawing the triangle.



One solution:

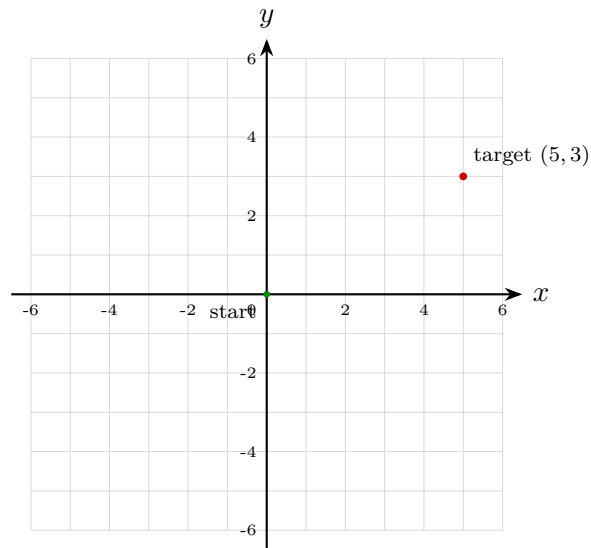
```
Start at (-3, -2) facing up, pen down.  
go to x: -3 y: -2  
go to x: 3 y: -2  
go to x: 0 y: 3  
go to x: -3 y: -2
```



Problem 7.

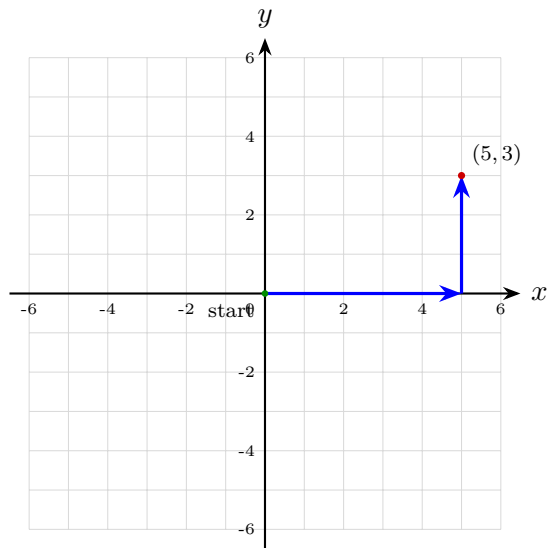
Instruct the turtle. The turtle is at the origin. Write `change x by` and `change y by` instructions to move it to the target.

change adds to where the turtle already is. How far across is the target, and how far up?



One solution:

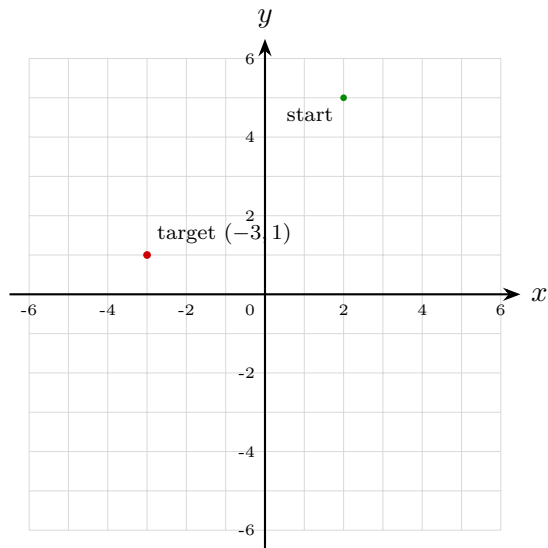
change x by 5
change y by 3



Problem 8.

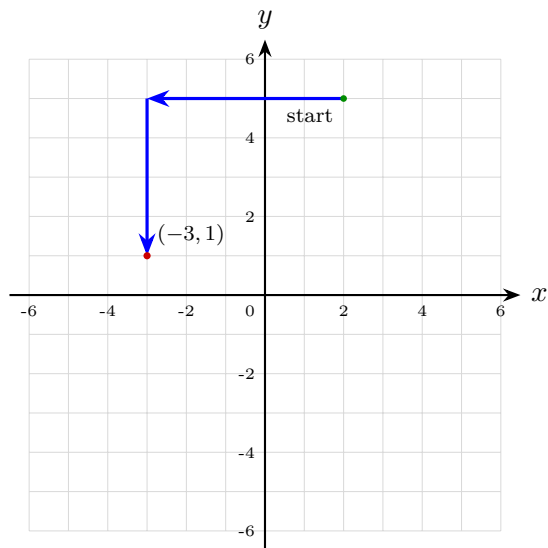
Instruct the turtle. The turtle starts at $(2, 5)$. Write **change** instructions to move it to the target. You will need negative numbers.

To move left, change x by a negative number; to move down, change y by a negative number.



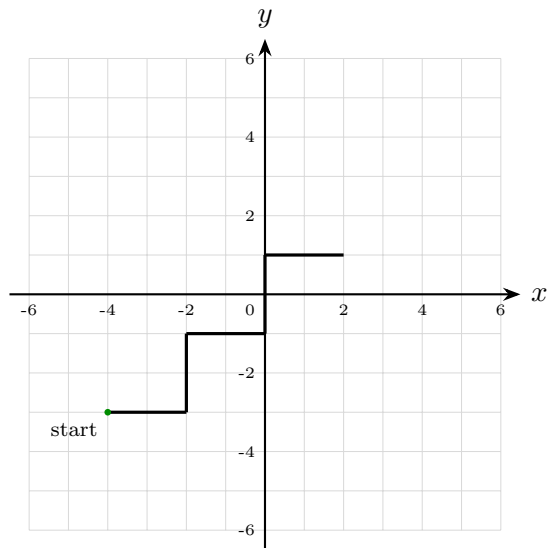
One solution:

Start at (2,5) facing up, pen down.
 change x by -5
 change y by -4



Problem 9.

Instruct the turtle. The turtle starts at $(-4, -3)$ with its pen down. Write **change x by** / **change y by** instructions to draw this staircase.



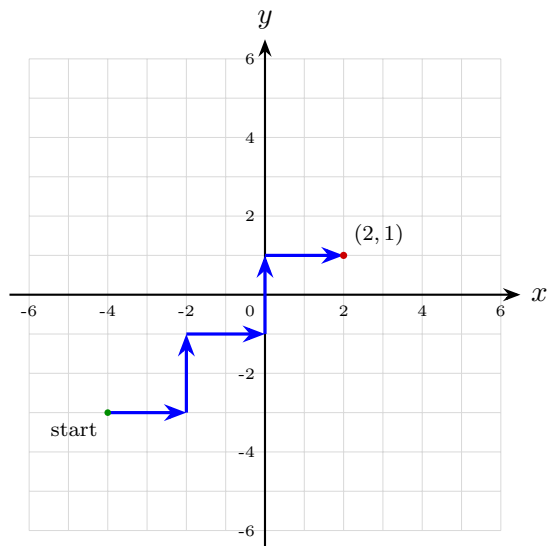
One solution:

Start at $(-4, -3)$ facing up, pen down.

```

change x by 2
change y by 2
change x by 2
change y by 2
change x by 2

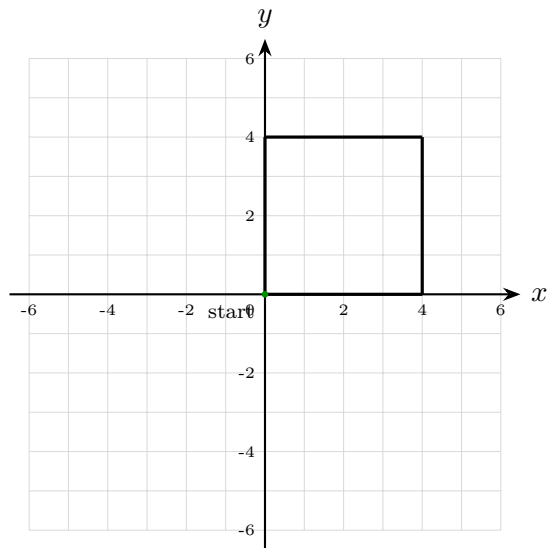
```



Problem 10.

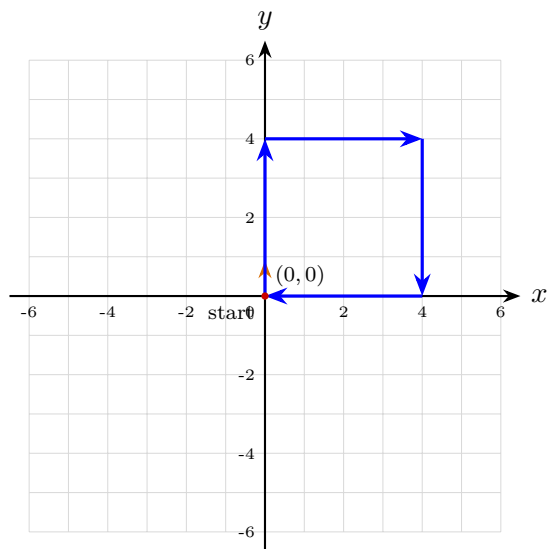
Instruct the turtle. This time use `move` and `turn` instead of `go to`. The turtle starts at the origin facing up. Write the instructions to draw this square.

move goes forward in whatever direction the turtle is facing; turn changes that direction without drawing.



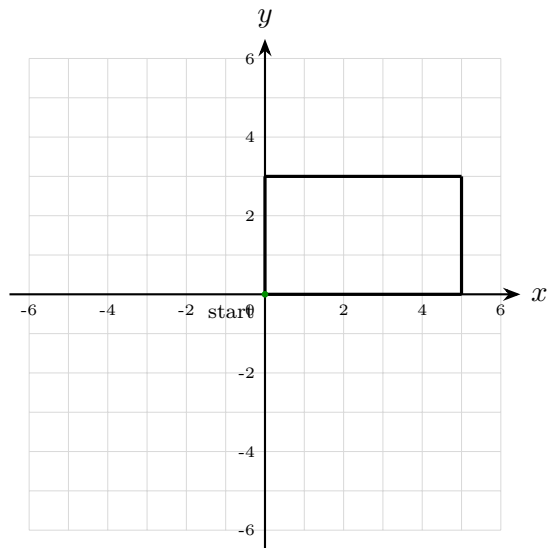
One solution:

```
move 4 steps
turn right 90 degrees
move 4 steps
turn right 90 degrees
move 4 steps
turn right 90 degrees
move 4 steps
```



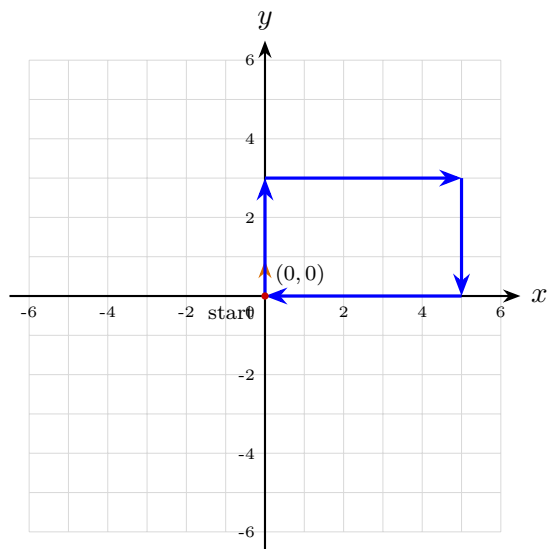
Problem 11.

Instruct the turtle. Using move and turn, draw this rectangle (5 across, 3 up). The turtle starts at the origin facing up.



One solution:

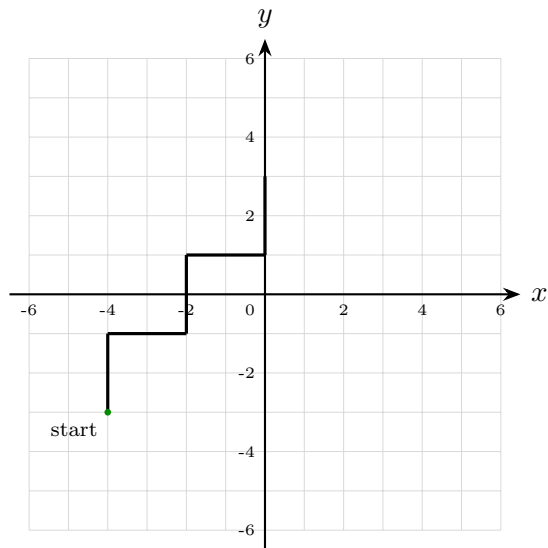
```
move 3 steps
turn right 90 degrees
move 5 steps
turn right 90 degrees
move 3 steps
turn right 90 degrees
move 5 steps
```



Problem 12.

Instruct the turtle. Using move and turn, draw these steps. The turtle starts at $(-4, -3)$ facing up. Watch which way it must turn at each corner.

Turning left then right then left keeps the turtle climbing in steps rather than closing a box.

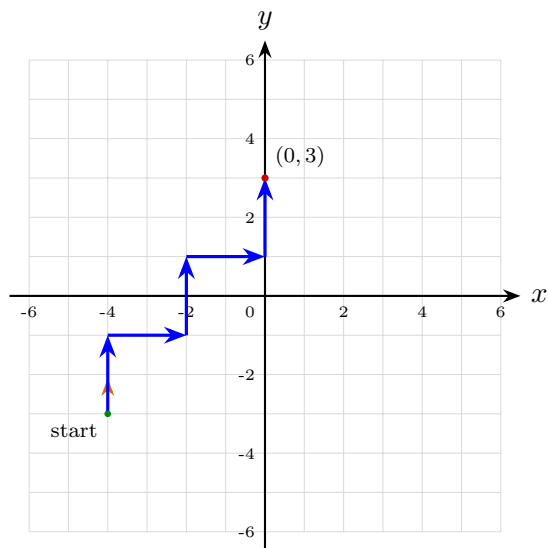


One solution:

```

Start at  $(-4, -3)$  facing up, pen down.
move 2 steps
turn right 90 degrees
move 2 steps
turn left 90 degrees
move 2 steps
turn right 90 degrees
move 2 steps
turn left 90 degrees
move 2 steps

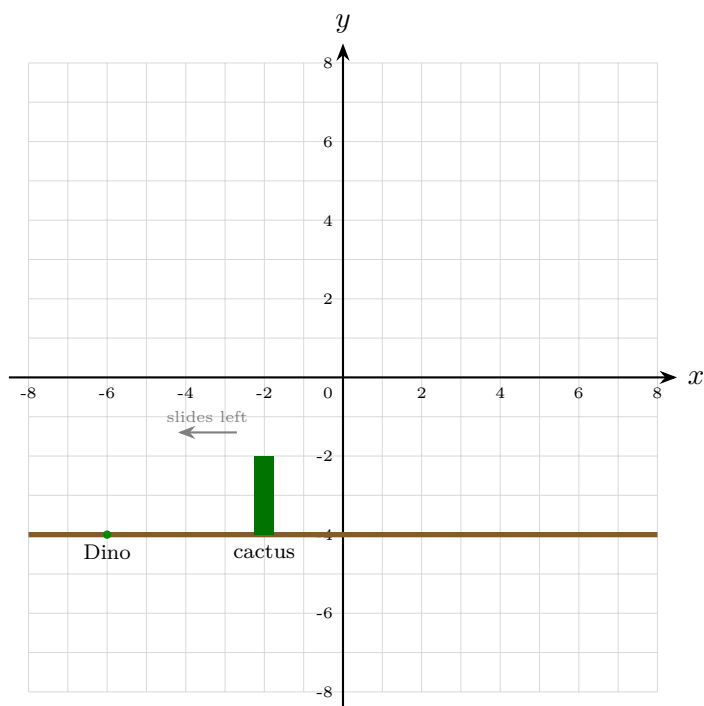
```



Problem 13.

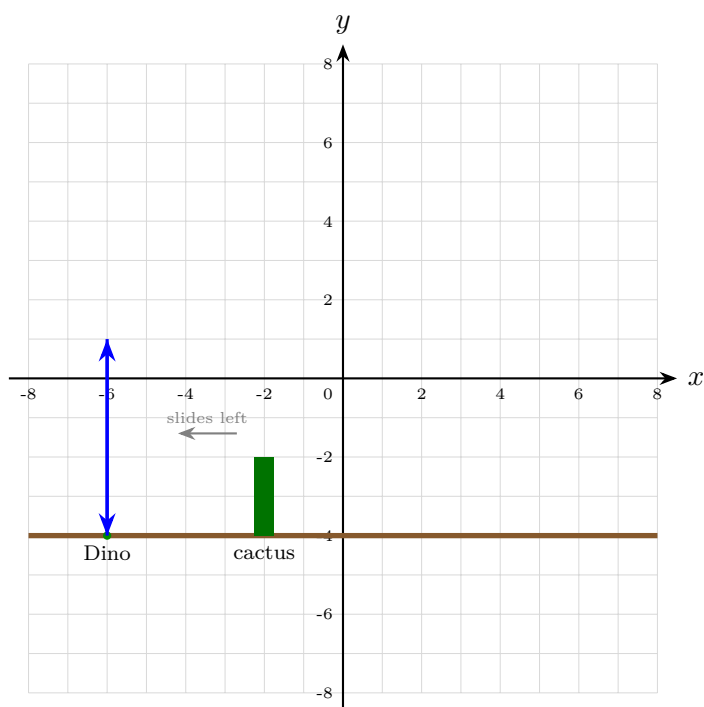
Instruct the turtle. Here is a mini Scratch stage (1 square = 20 steps). The Dino rests on the floor; a cactus slides in from the right. Write the **change y** by instructions for the Dino's jump: go up 5, then come back down 5 to land on the floor.

The Dino never moves across — it only changes y . The cactus passes underneath using *change x by a negative number*.



One solution:

Start at $(-6, -4)$ facing up, pen down.
change y by 5
change y by -5



*In the real game the floor is at $y = -80$ and the cactus spawns at $x = 240$. The Dino stays at one x and only **change y by** to jump, while the cactus slides left with **change x by**. You just wrote the heart of the game.*
